

WIND POWER: The Global Impact



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"Of all the forces of nature, I should think the wind contains the largest amount of motive power—that is, power to move things. Take any given space of the earth's surface—for instance, Illinois; and all the power exerted by all the men, and beasts, and running-water, and steam, over and upon it, shall not equal the one hundredth part of what is exerted by the blowing of the wind over and upon the same space. And yet it has not, so far in the world's history, become proportionally valuable as a motive power. It is applied extensively, and advantageously, to sail-vessels in navigation. Add to this a few windmills, and pumps, and you have about all. ...As yet, the wind is an untamed, and unharnessed force; and quite possibly one of the greatest discoveries hereafter to be made will be the taming and harnessing of it."

—Abraham Lincoln

Globally, there is a huge emphasis on encashing the energy generation possibilities of renewable sources, especially wind. Wind power offers a sustainable option in the pursuit of renewable energy. In this article we explore how wind energy is being utilised in different parts of the world.

A case study from Canada

Wind power promises to be more beneficial than any other existing source of energy. It doesn't produce any undesirable and harmful waste, harnesses the power of nature (wind), and is one of the best alternatives to burning of coal/gas or use of nuclear energy.

The Wolfe Island Wind Project in Canada aimed at meeting electricity needs of residents in Kingston, Ontario, through wind power. As part of the project, 86 turbines were placed in the wind farm with an investment of around 410 million dollars to generate sufficient electricity for 75,000 homes. The returns for the same were spread over 20-25 years.

Costs and benefits. Turbine generators for the project cost 1500-2000 dollars per kilowatt. Using 86 turbines of 2.3MW capacity with a total cost of 410 million dollars, the project cost a little more than 2000 dollars per kilowatt.

The wind power cost eight cents to 10.2 cents per kilowatt hour, which was much higher than the actual electricity price in Canada (1.2 to 7.8 cents per kilowatt hour, depending upon the location). On adding

